Isomo Pulse Starfleet Monitor - System Requirements

Document date: April 26, 2026

1. Purpose

Starfleet Monitor provides operational visibility for Isomo school connectivity. It collects laptop health, local network quality, Starlink telemetry, and site metadata so administrators can understand which schools are online, which devices are healthy, and where intervention is needed.

2. System Scope

The system includes:

Component	Scope
Windows agent	Collects laptop, latency, usage, and Starlink telemetry from managed laptops
Backend API	Receives ingest payloads, authenticates users/agents, resolves sites, stores data, and serves dashboard/mobile clients
Database	Stores sites, devices, telemetry, usage, health, scores, users, and site change events
Dashboard	Provides operations visibility for sites, Starlinks, laptops, charts, and alerts
Mobile app	Provides field/admin companion access
Intune deployment	Installs and updates the Windows agent through remediation scripts

Out of scope for this document:

Item	Notes
Starlink account billing management	The platform observes service health but does not manage Starlink billing
Intune tenant governance	The platform provides scripts and guidance, but tenant policy ownership remains with the admin
Physical installation	Dish placement, power, cabling, and school network layout are operational responsibilities

3. Users And Roles

Role	Responsibilities
Admin	Manage platform access, generate agent tokens, review all sites, acknowledge issues
Operations user	Monitor school health, investigate failures, act on site changes and stale laptops
Field user	Review site/device status from the mobile app during support visits

Role	Responsibilities
Managed laptop	Runs the agent as SYSTEM through Windows Task Scheduler

4. Functional Requirements

4.1 Site And Starlink Inventory

ID	Requirement	Priority	Status
FR-001	Store school/site records with name, location, district, coordinates, Starlink serial, Starlink UUID, and kit ID	Must	Implemented
FR-002	Return site inventory through authenticated API endpoints	Must	Implemented
FR-003	Support import/update from the Starlink management master sheet	Must	In progress
FR-004	Preserve stable internal site IDs while also storing master sheet IDs	Should	Implemented

4.2 Laptop Agent

ID	Requirement	Priority	Status
FR-010	Run automatically on Windows laptops every 5 minutes	Must	Implemented
FR-011	Report heartbeat with device identity, hostname, OS, model, and manufacturer	Must	Implemented
FR-012	Report battery, disk, and memory health	Must	Implemented
FR-013	Report ping latency percentiles to a configured host	Must	Implemented
FR-014	Report daily network usage deltas	Must	Implemented
FR-015	Queue failed ingest payloads and replay later without double-counting usage	Must	Implemented
FR-016	Report agent health including version, queue depth, last error, and run ID	Must	Implemented
FR-017	Distinguish Intune-managed installs from manual/legacy installs	Must	Implemented

4.3 Starlink Telemetry

ID	Requirement	Priority	Status
FR-020	Read local Starlink gRPC-web diagnostics when `192.168.100.1:9201` is reachable	Must	Implemented
FR-021	Extract GPS coordinates, azimuth, and elevation from diagnostics byte offsets when protobuf decoding is incomplete	Must	Implemented
FR-022	Extract raw Starlink terminal ID in `ut...` format when available	Must	Implemented
FR-023	Send both raw `starlink_id` and normalized `starlink_uuid` to backend	Must	Implemented
FR-024	Continue laptop heartbeat/usage/latency ingest even when Starlink telemetry is unavailable	Must	Implemented

4.4 Site Resolution

ID	Requirement	Priority	Status
FR-030	Resolve site from Starlink UUID inventory when terminal identity is available	Must	Implemented
FR-031	Resolve site from GPS proximity when valid coordinates are available	Must	Implemented
FR-032	Fall back to configured `site_id` when identity and GPS are unavailable	Must	Implemented
FR-033	Avoid false site moves using movement thresholds and event logging	Should	Implemented

4.5 Dashboard And Mobile

ID	Requirement	Priority	Status
FR-040	Show site inventory, online laptop counts, signal metrics, score, and usage	Must	Implemented
FR-041	Show laptop list with health and stale status	Must	Implemented
FR-042	Show site detail with connected devices and latest signal	Must	Implemented
FR-043	Show historical latency, signal, and usage charts	Should	In progress
FR-044	Support mobile field/admin workflows	Should	In progress

4.6 Authentication And Authorization

ID	Requirement	Priority	Status
FR-050	Authenticate dashboard users through `/auth/login`	Must	Implemented
FR-051	Require signed tokens for API access	Must	Implemented
FR-052	Generate site-scoped agent tokens for Intune deployment	Must	Implemented
FR-053	Reject agent ingest when token site scope does not match posted site hint	Must	Implemented
FR-054	Avoid embedding admin tokens in agent or Intune scripts	Must	Required operating practice

5. Non-Functional Requirements

Category	Requirement
Availability	Backend should remain available for laptop ingest and dashboard reads during normal school operations
Offline tolerance	Agent must preserve data locally while backend or network is unavailable
Idempotency	Retried payloads must not double-count usage metrics
Security	Production secrets, JWTs, and generated Intune scripts with real tokens must not be committed
Observability	Agent logs and backend health endpoint must provide enough signal for field diagnosis
Maintainability	Migrations must be versioned and run automatically in production deploys
Portability	Agent must run under Windows PowerShell 5.1 as SYSTEM
Privacy	Store operational telemetry required for fleet support, not user content

6. Infrastructure Requirements

6.1 Production Services

Service	Requirement
Neon PostgreSQL	PostgreSQL database reachable through `DATABASE_URL` with SSL
Railway	Node.js 20 or newer backend runtime with environment variables configured
Vercel	Dashboard deployment with API base URL and allowed CORS origin configured

Service	Requirement
Microsoft Intune	Windows Remediation package assigned to test and production device groups

6.2 Backend Runtime

Requirement	Value
Node.js	20 or newer
Package manager	npm/yarn compatible workspace install
Start command	`npm start` in `packages/backend`
Migration command	`node migrate.js`
Health check	`GET /health` returns API and DB status

6.3 Windows Agent Runtime

Requirement	Value
OS	Windows 10/11 managed by Intune
PowerShell	Windows PowerShell 5.1
Execution context	SYSTEM
Install directory	`C:\ProgramData\Starfleet`
Scheduled task	`StarfleetPulse`
Frequency	Every 5 minutes
Network	Access to backend API and local Starlink dish IP when on site

7. Data Requirements

Data set	Required fields
Sites	ID, master sheet ID, name, district, location, latitude, longitude, Starlink UUID, Starlink serial, kit ID
Devices	Windows serial/UUID, hostname, OS, model, manufacturer, site ID, last seen, last ingest success
Signal readings	Site ID, device ID, timestamp, latency, SNR, obstruction, drop rate, throughput, GPS
Usage readings	Device ID, date, bytes down, bytes up, payload identity
Health readings	Device ID, timestamp, battery, disk, RAM
Agent health	Device ID, version, queue depth, last error, run ID, last success
Site changes	Device ID, from site, to site, distance, evidence, acknowledgement

8. Deployment Requirements

8.1 Backend

1. Merge backend changes to the Railway-tracked branch.
2. Railway runs migrations and starts the API.
3. Verify `GET https://api.starfleet.icircles.rw/health`.
4. Verify `/api/agent-tokens` returns `401 Unauthorized` without auth and works with admin auth.

8.2 Dashboard

1. Merge frontend/shared changes to the Vercel-tracked branch.
2. Confirm the dashboard uses the production API base URL.
3. Verify login, sites, laptops, and site detail screens.

8.3 Agent

1. Generate a site-scoped agent token through the backend.
2. Generate the Intune remediation upload file.
3. Paste the agent token into the generated remediation file.
4. Upload `packages/agent/detection.ps1` and `dist/intune/remediation.ps1`.
5. Configure Intune to run as SYSTEM in 64-bit PowerShell.
6. Assign to the test VM before broad deployment.
7. Confirm `install_source=intune_remediation`, agent version, heartbeat, and no `401` ingest errors.

9. Acceptance Criteria

Area	Acceptance criteria
Agent install	Intune creates `C:\ProgramData\Starfleet`, writes config, writes install marker, creates scheduled task
Agent ingest	Heartbeat succeeds and updates `last_heartbeat.txt`; queue drains to zero
Auth	Agent token is site-scoped and accepted by ingest endpoints; admin login token is not used by agent
Starlink identity	Backend maps `ut31...` to stored UUID `31...` and resolves the correct site
GPS fallback	Backend can resolve site from coordinates when identity is missing
Dashboard	Operators can see current sites, laptops, status, usage, and alerts
Production health	Railway health endpoint reports API and DB healthy

10. Key Risks

Risk	Impact	Mitigation
Wrong token embedded in Intune script	Agent receives `401 Unauthorized` and cannot ingest	Generate site-scoped agent token, verify logs before broad rollout
Starlink GPS disabled	GPS-based site inference stops working	Use Starlink UUID inventory as primary identity fallback
Separate Intune platform script remains assigned	Old and new installers can conflict	Use one remediation package and unassign legacy script during testing

Risk	Impact	Mitigation
Generated token-bearing script committed	Credential exposure	Keep `dist/` ignored and review git status before commits
Local dish API unreachable from laptop	Starlink metrics missing for that cycle	Continue heartbeat/health/usage ingest and rely on next cycle

11. Open Decisions

Decision	Owner	Notes
Agent token rotation interval	Admin/Ops	Current default is long-lived site token; define rotation policy
Production Intune grouping	Admin/Ops	Decide school-by-school rollout groups after VM validation
Notification recipient model	Admin/Ops	Confirm email/push recipients and acknowledgement workflow
Dashboard public URL	Admin/Ops	Confirm final Vercel domain and CORS entry